Applied Mechanics Workbook

A Concise Introduction to Computational Tools

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Preface

A Concise Introduction to Computational Tools provides a series of tutorials derived from lecture notes, offering clear and focused guidance on essential topics. No prior programming experience is required.

Students are expected to be comfortable with college-level mathematics and science and are strongly encouraged to consult reliable reference texts. For example, foundational and practical texts in mathematics, applied mechanics and computational tools, including Bird (2021), Hannah & Hillier (1995), Russell et al. (2021), Bolton (2021), Shaw (2017), Sweigart (2021), and Sweigart (2020).

The tutorials also assume working familiarity with either macOS or Microsoft Windows. For best results, these materials should be used on a laptop or desktop computer rather than a tablet or smartphone.

Chapter 1

Units, Vectors, and Scalar Quantities

1.1 SI Units

The book adopts the **Système International (SI)** of units. The base units most relevant to mechanics are:

Quantity	Unit	Symbol
Mass	kilogram	kg
Length	metre	m
Time	second	s
Force	newton	N
Energy / Work	joule	J
Power	watt	W
Pressure / Stress	pascal	Pa

Derived units follow from these. For example:

$$1\ \mathrm{N} = 1\ \mathrm{kg} \cdot \mathrm{m/s}^2 \qquad 1\ \mathrm{Pa} = 1\ \mathrm{N/m}^2 \qquad 1\ \mathrm{J} = 1\ \mathrm{N} \cdot \mathrm{m}$$

1.2 Scalar and Vector Quantities

Scalar quantities have magnitude only and obey ordinary arithmetic:

- Examples: mass, speed, energy, temperature, time

Vector quantities have both magnitude and direction and must be combined using vector algebra:

- Examples: force, velocity, acceleration, displacement, momentum

A vector is represented graphically as an arrow — its length denotes magnitude, its orientation denotes direction.

1.3 Vector Addition

1.3.1 Triangle Rule

Two vectors **A** and **B** are added tip-to-tail; the resultant **R** closes the triangle:

$$\mathbf{R} = \mathbf{A} + \mathbf{B}$$

1.3.2 Parallelogram Rule

Both vectors are drawn from the same point; the resultant is the diagonal of the parallelogram they form.

1.3.3 Polygon Rule

For more than two vectors, they are placed tip-to-tail in sequence; the resultant closes the polygon from the tail of the first to the tip of the last.

1.4 Vector Subtraction

Subtracting **B** from **A** is equivalent to adding the negative of **B**:

$$\mathbf{R} = \mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$$

1.5 Resolution of Vectors

Any vector **F** can be resolved into two perpendicular components. For a vector at angle θ to the horizontal:

$$F_x = F\cos\theta$$

$$F_y = F\sin\theta$$

This is the most widely used technique in the book — resolving all forces into horizontal and vertical components before applying equilibrium conditions.

1.6 Resultant of Several Vectors

To find the resultant of a system of vectors analytically:

1. Resolve each vector into x and y components
2. Sum all components in each direction:

$$\Sigma F_x = F_{1x} + F_{2x} + \cdots \qquad \Sigma F_y = F_{1y} + F_{2y} + \cdots$$

3. Find the magnitude of the resultant:

$$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$$

4. Find the direction:

$$\theta = \arctan\left(\frac{\Sigma F_y}{\Sigma F_x}\right)$$

1.7 The Laws of Sines and Cosines

For cases involving triangles of vectors, these laws provide an analytical alternative to graphical methods.

Sine rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

Cosine rule:

$$c^2 = a^2 + b^2 - 2ab\cos C$$

1.8 Summary of Equations

Concept	Equation
Horizontal component	$F_x = F\cos\theta$
Vertical component	$F_y = F\sin\theta$
Resultant magnitude	$R = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2}$
Resultant direction	$\theta = \arctan(\Sigma F_y/\Sigma F_x)$
Cosine rule	$c^2 = a^2 + b^2 - 2ab\cos C$

1.9 Summary

The resolution-and-recombination technique — breaking every vector into perpendicular components, summing each direction separately, then reconstructing the resultant — is the single most important skill introduced in this chapter. It underpins equilibrium analysis, dynamics, stress transformation, and virtually every other topic in the book.

Chapter 2

Balancing

2.1 Newton's Laws of Motion

2.1.1 First Law — Inertia

> *A body remains at rest or in uniform straight-line motion unless acted upon by a net external force.*

A body in motion travels in a straight line unless forced to change direction. This resistance to any change in motion is called **inertia**. The law defines what a force *is*: the only thing capable of altering a body's state of motion.

$$\sum \vec{F} = 0 \implies \vec{v} = \text{constant}$$

2.1.2 Second Law — Force and Acceleration

> *The net force acting on a body equals its mass multiplied by its acceleration.*

$$\boxed{F = ma}$$

Force and acceleration act in the same direction. The relationship is linear: doubling the net force doubles the acceleration; doubling the mass halves it for the same force.

Change	Effect on acceleration
Force doubled	Acceleration doubled
Mass doubled	Acceleration halved
Both doubled	Acceleration unchanged

2.1.3 Third Law — Action and Reaction

> *For every force one body exerts on another, the second body exerts an equal and opposite force back on the first.*

$$\vec{F}_{A\to B} = -\vec{F}_{B\to A}$$

Key points:

- The forces are **equal in magnitude** and **opposite in direction**.
- They act on **different objects** — they are never a cancelling pair.
- The reaction force has the same nature as the action (both contact, both gravitational, etc.).

2.2 Centrifugal and Centripetal Forces

A body in motion travels in a straight line unless forced to change direction (Newton's First Law). When moving in a circle, its inertia causes it to resist that continuous change of direction — in a rotating reference frame, this resistance appears as a fictitious outward "centrifugal" force. The real inward force that continuously deflects the body onto the circular path is the **centripetal force**. Because circular motion involves a constant change of direction, velocity is always changing even at constant speed. For small angles, the magnitude of this change in velocity simplifies to $v\theta$.

2.3 Static Balance

A system is statically balanced when the resultant of all centrifugal forces due to rotating masses is zero — i.e., the centre of mass lies on the axis of rotation. Static balance requires the vector sum of all mr terms to be zero:

$$\sum mr = 0$$

A single correction mass placed in the same plane can achieve static balance.

2.4 Dynamic Balance

Static balance alone is insufficient for systems with masses distributed across different axial planes. Dynamic balance requires two conditions to be satisfied simultaneously:

1. The vector sum of all centrifugal forces equals zero (static condition).
2. The vector sum of all **moments** of these forces about any reference plane equals zero.

$$\sum mr = 0 \qquad \text{and} \qquad \sum mr\ell = 0$$

This generally requires **two correction masses** placed in two chosen planes.

2.5 Balancing in a Single Plane

For coplanar rotating masses, a graphical or analytical approach uses the **force polygon**. Closing the mr vector polygon determines the required balancing mass and its angular position.

2.6 Balancing in Several Planes

For masses distributed along a shaft, the procedure is:

1. Choose a reference plane and construct a **couple polygon** using $mr\ell$ terms, where ℓ is the axial distance from the reference plane.
2. Closing the couple polygon gives the first correction mass (magnitude and angle).
3. Construct the **force polygon** using mr terms to find the second correction mass.

2.7 Balancing of Reciprocating Machinery

A reciprocating engine generates forces that are fundamentally harder to balance than rotating masses. The piston moves in one plane only — adding an opposite mass in that plane creates a new out-of-balance force perpendicular to it. The problem cannot be solved by a single counterweight alone.

2.7.1 Primary and Secondary Forces

Treating the connecting rod as two lumped masses — one at the crank pin, one at the gudgeon pin — splits the system into a rotating part and a pure reciprocating part. The total reciprocating force is:

$$F = mr\omega^2\left(\cos\theta + \frac{r}{\ell}\cos 2\theta + \cdots\right)$$

Term	Name	Frequency
$\cos\theta$	Primary force	ω (crank speed)
$\frac{r}{\ell}\cos 2\theta$	Secondary force	2ω
Higher terms	Negligible in practice	—

The secondary force arises from the obliquity of the connecting rod; higher harmonics are generally ignored. Written separately:

Primary force (first-order):

$$F_{\text{primary}} = mr\omega^2 \cos\theta$$

Secondary force (second-order):

$$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$$

where $n = \ell/r$ is the ratio of connecting rod length ℓ to crank radius r.

2.8 Methods for Complete Balancing

2.8.1 Crankshaft Counterweight (Partial Balance)

Adding a mass opposite the crank pin cancels the rotating component entirely but only reduces the primary reciprocating force — it cannot eliminate it. This trades a vertical unbalance for a smaller horizontal one. A typical compromise balances **50–60%** of the reciprocating mass.

2.8.2 Multi-Cylinder In-Line Engines

By phasing cranks at equal angles, primary and secondary forces and couples can cancel across the engine. Balance requires:

$$\sum mr = 0 \quad \text{and} \quad \sum mra = 0$$

for both primary and secondary components.

Configuration	Primary F	Primary C	Secondary F	Secondary C
2-cyl, 180°				
4-cyl, 180°				
6-cyl, 60°				

The **six-cylinder in-line engine** achieves complete balance for both primary and secondary effects.

2.8.3 Partial Balance

Complete balance is not always possible or practical — particularly for single-cylinder engines. In such cases a balance mass is added to reduce the dominant unbalanced force, and a residual force in one direction is accepted to eliminate the more harmful force in the perpendicular direction.

2.9 Critical Speed

Every rotating shaft is also an elastic beam with natural frequencies of lateral vibration. The **critical speed** is the rotational speed at which the excitation frequency from residual unbalance equals a natural frequency, producing resonance.

For a shaft carrying a disc, using the static deflection δ_{st} under its own load:

$$\boxed{N_c = \frac{60}{2\pi}\sqrt{\frac{g}{\delta_{st}}}}$$

This is convenient because δ_{st} can be directly measured or calculated from beam theory.

2.9.1 Behaviour Around Critical Speed

Speed regime	Shaft behaviour
Well below N_c	Shaft runs stiffly; centre of mass geometric centre
Approaching N_c	Amplitude grows rapidly — whirling begins
At N_c	Resonance; amplitude theoretically unbounded (damping limits it)
Well above N_c	Shaft self-centres; centre of mass migrates toward axis

Many turbines and high-speed spindles operate **supercritically** — above the first critical speed — and simply pass through it during run-up.

2.9.2 Multiple Critical Speeds

A real shaft has infinitely many lateral modes, each giving a critical speed:

$$N_{c_n} \propto n^2 \quad \text{(uniform simply-supported shaft)}$$

The **first critical speed** is the most dangerous: it is encountered first and produces the largest deflection amplitude.

2.10 Summary of Equations

Concept	Equation
Centrifugal force	$F = mr\omega^2$
Static balance	$\sum mr = 0$
Dynamic balance	$\sum mr = 0$ and $\sum mr\ell = 0$
Primary reciprocating force	$F_{\text{primary}} = mr\omega^2 \cos\theta$
Secondary reciprocating force	$F_{\text{secondary}} = \frac{mr\omega^2}{n} \cos 2\theta$
Critical speed	$N_c = \frac{60}{2\pi}\sqrt{\frac{g}{\delta_{st}}}$

2.11 Summary

Balancing is fundamentally a **vector problem**. Both graphical (polygon) and analytical methods are used throughout. Dynamic balancing always requires satisfying both force *and* moment equilibrium, building progressively from simple single-plane cases up to the complexity of multi-cylinder engine analysis. Critical speed analysis adds a further dimension: even a perfectly balanced shaft can experience destructive resonance if its operating speed coincides with a lateral natural frequency.

Chapter 3

Simple Harmonic Motion

Simple Harmonic Motion: The foundation of vibration analysis. A particle undergoes SHM when its acceleration is proportional to displacement and directed toward the equilibrium position:

$$a = -\omega^2 x$$

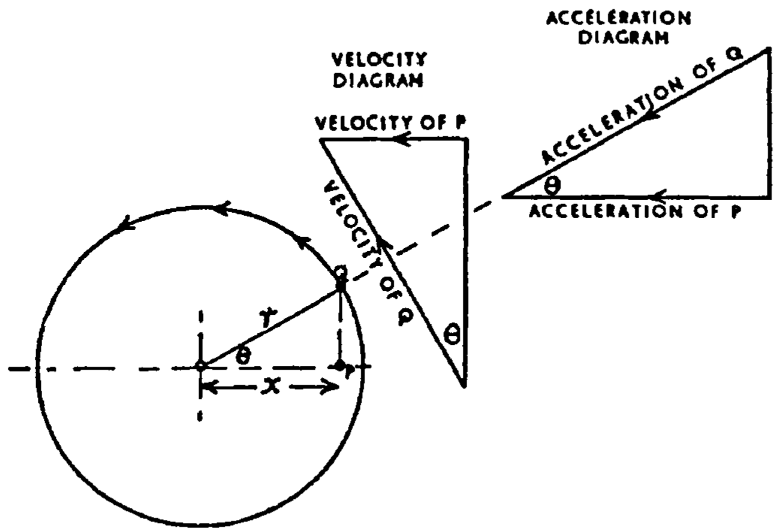
Key relationships such as displacement, velocity, acceleration, amplitude, period, and frequency are all derived from this condition.

Simple harmonic motion is a particular form of reciprocating, or 'to-and-fro', motion in which the acceleration and the velocity of the body vary as it moves from one end of its travel to the other; the important characteristic of this motion is that the acceleration is proportional to the displacement from mid-travel (and directed towards it).

When a body moves forwards and backwards with simple harmonic motion, at one end of its oscillation the body is momentarily at rest; it receives maximum acceleration, which causes it to move with increasing velocity. The acceleration, which is proportional to the displacement from mid-travel, decreases, and its velocity increases as the body approaches mid-travel. As it passes its mid-position the acceleration is zero and the velocity is maximum. From mid-travel onwards the acceleration is negative and increasing in magnitude while the velocity decreases until, at the other end of its travel, the velocity is momentarily zero and the acceleration is maximum to return the body in the opposite direction.

If a particle is travelling around a circular path in a vertical plane at a constant speed and its shadow could be seen on a vertical screen, the shadow would move forwards and backwards with simple harmonic motion.

From a practical point of view, neglecting the effect of the angularity of the connecting rod of a reciprocating engine, the piston moves with simple harmonic motion when the crank pin travels at a constant angular velocity.



Referring to Fig. the circle represents the circular path of a particle Q moving at a constant velocity; this could be the crank pin of an engine. The projection of Q on to the plane of the diameter is point P, which could represent the relative position and motion of the piston (neglecting angularity of the connecting rod). Q is moving at a constant angular velocity of ω around the circular path of radius r, so its linear velocity is ωr. The velocity vector diagram is drawn to represent the linear velocity of Q when passing the point at θ past dead centre; the horizontal component represents the velocity of P at that instant. Thus:

$$\text{Velocity of } P = \text{Velocity of } Q \times \sin\theta = \omega r \sin\theta$$

The angular velocity ω is constant; therefore the acceleration of a body moving with simple harmonic motion is proportional to its displacement from mid-travel (and directed towards it).

The **AMPLITUDE** is the maximum displacement to either side of mid-travel.

The **PERIODIC TIME** is the time taken to make one complete oscillation, that is, to move completely from one end to the other and back again.

The time for P to make one complete oscillation is the same time that Q takes to complete one full revolution. Since the distance travelled in one revolution is $2\pi r$ and the linear velocity of Q is ωr:

$$\text{Periodic time} = \frac{\text{distance}}{\text{velocity}} = \frac{2\pi r}{\omega r} = \frac{2\pi}{\omega}$$

Since the acceleration of P is proportional to its displacement from mid-travel:

$$\text{acceleration} = \omega^2 \times \text{displacement}$$

it follows that:

$$\omega = \sqrt{\frac{\text{acceleration}}{\text{displacement}}}$$

Substituting into the expression for periodic time:

$$\text{Periodic time} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{\text{acceleration}}{\text{displacement}}}}$$

This expression gives the periodic time for any body moving with simple harmonic motion.

The **FREQUENCY** is the number of complete oscillations made in one second.

If t = periodic time (s), then:

$$\text{Frequency} = \frac{1}{t} \quad \text{oscillations per second}$$

3.1 Free Undamped Vibration

3.1.1 Mass-Spring System

The simplest model: a mass m on a spring of stiffness k:

$$\omega_n = \sqrt{\frac{k}{m}} \qquad f_n = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

3.1.2 Simple Pendulum

$$f_n = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$$

3.1.3 Equivalent Stiffness

Springs in series and parallel are reduced to a single equivalent stiffness before applying the standard formulae:

$$k_{\text{series}} = \frac{k_1 k_2}{k_1 + k_2} \qquad k_{\text{parallel}} = k_1 + k_2$$

3.2 Energy Method (Rayleigh's Method)

An alternative to force methods — equating maximum kinetic energy to maximum potential energy to find the natural frequency. Particularly useful where writing equations of motion is complex:

$$\frac{1}{2}mv_{\max}^2 = \frac{1}{2}kx_{\max}^2$$

3.3 Damped Free Vibration

In practice, energy is dissipated. A viscous damping force proportional to velocity is assumed:

$$F_d = -c\dot{x}$$

The **damping ratio** ζ determines the character of the motion:

$$\zeta = \frac{c}{2\sqrt{mk}}$$

Condition	Behaviour
$\zeta < 1$	Underdamped — oscillates with decaying amplitude
$\zeta = 1$	Critically damped — returns to rest without oscillation
$\zeta > 1$	Overdamped — sluggish return to rest

The **damped natural frequency** is reduced from the undamped value:

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

3.4 Forced Vibration (Undamped)

When a harmonic exciting force $F = F_0\sin(\omega t)$ is applied, the steady-state amplitude is:

$$X = \frac{F_0/k}{1-(\omega/\omega_n)^2}$$

Note that above resonance this expression yields a negative value, indicating that the response is 180° out of phase with the excitation; the physical amplitude is the modulus $|X|$.

Resonance occurs when $\omega \to \omega_n$, causing theoretically infinite amplitude.

3.5 Forced Vibration (Damped)

Damping limits the amplitude at resonance. The **dynamic magnification factor (DMF)** gives the ratio of dynamic to static deflection:

$$\text{DMF} = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_n}\right)^2\right]^2 + \left[2\zeta\frac{\omega}{\omega_n}\right]^2}}$$

At resonance ($\omega = \omega_n$), the DMF reduces to:

$$\text{DMF}_{\text{resonance}} = \frac{1}{2\zeta}$$

3.6 Whirling of Shafts

A rotating shaft has a **critical speed** at which it deflects violently — directly analogous to resonance. The critical (whirling) speed corresponds to the natural frequency of transverse vibration:

$$N_c = \frac{30}{\pi}\sqrt{\frac{g}{\delta_{st}}}$$

where δ_{st} is the static deflection of the shaft under its own (or the rotor's) weight. Operating speeds should be well clear of N_c.

3.7 Summary of Equations

Concept	Equation
SHM condition	$a = -\omega^2 x$
Natural frequency (spring-mass)	$\omega_n = \sqrt{k/m}$
Natural frequency (pendulum)	$f_n = \frac{1}{2\pi}\sqrt{g/L}$
Damping ratio	$\zeta = c/(2\sqrt{mk})$
Damped natural frequency	$\omega_d = \omega_n\sqrt{1-\zeta^2}$
Resonance condition	$\omega = \omega_n$
DMF (damped forced)	$1/\sqrt{[1-r^2]^2 + [2\zeta r]^2}$, where $r = \omega/\omega_n$
Critical shaft speed	$N_c = (30/\pi)\sqrt{g/\delta_{st}}$

3.8 Summary

All vibration problems reduce to understanding the interplay between **stiffness** (restoring force), **mass** (inertia), and **damping** (energy dissipation). Resonance avoidance and the role of damping in controlling amplitudes are the central practical concerns for mechanical design.

Chapter 4

Stress & Strain

4.1 Direct Stress and Strain

When a bar is subjected to an axial force F over cross-sectional area A, the direct stress is:

$$\sigma = \frac{F}{A}$$

The resulting axial strain is the ratio of extension to original length:

$$\varepsilon = \frac{\delta L}{L}$$

4.2 Young's Modulus (Modulus of Elasticity)

Within the elastic limit, stress and strain are proportional — **Hooke's Law**:

$$E = \frac{\sigma}{\varepsilon}$$

This gives the deformation of a bar under axial load:

$$\delta L = \frac{FL}{AE}$$

4.3 Shear Stress and Shear Strain

Shear stress acts tangentially across a plane:

$$\tau = \frac{F}{A}$$

Shear strain γ is the angular deformation (in radians). The **Modulus of Rigidity** (shear modulus) relates them:

$$G = \frac{\tau}{\gamma}$$

4.4 Poisson's Ratio

Axial loading causes lateral contraction (or expansion) as well as axial strain. Poisson's ratio ν relates the two:

$$\nu = -\frac{\varepsilon_{\text{lat}}}{\varepsilon_{\text{axial}}}$$

Typical values for metals: $\nu \approx 0.25$–0.33.

4.5 Volumetric Strain and Bulk Modulus

Under hydrostatic (equal all-round) stress, the volumetric strain is:

$$\varepsilon_v = \frac{\delta V}{V}$$

The **Bulk Modulus** K relates hydrostatic stress to volumetric strain:

$$K = \frac{\sigma}{\varepsilon_v}$$

4.6 Relationship Between Elastic Constants

The three elastic constants E, G, and K are not independent — they are linked through Poisson's ratio:

$$E = 2G(1 + \nu)$$

$$E = 3K(1 - 2\nu)$$

4.7 Composite Bars

When two materials are bonded together (e.g. a steel rod inside a brass tube), two equations are needed:

Compatibility — both materials deform by the same amount:

$$\delta L_1 = \delta L_2 \quad \Rightarrow \quad \frac{\sigma_1}{E_1} = \frac{\sigma_2}{E_2}$$

Equilibrium — internal forces sum to the applied load:

$$\sigma_1 A_1 + \sigma_2 A_2 = F$$

Solving these simultaneously gives the stress in each material.

4.8 Thermal Stress

If a bar is **prevented from expanding freely** when heated, a compressive stress is induced:

$$\sigma = E\alpha\Delta T$$

where:

- α = coefficient of linear thermal expansion
- ΔT = temperature change

For a composite bar with mismatched expansion, the same compatibility and equilibrium approach applies.

4.9 Factor of Safety

The factor of safety relates the failure stress to the permissible working stress:

$$\text{FoS} = \frac{\text{failure stress}}{\text{allowable (working) stress}}$$

A higher FoS reflects greater uncertainty in loading or material properties.

4.10 Summary of Equations

Concept	Equation
Direct stress	$\sigma = F/A$
Direct strain	$\varepsilon = \delta L/L$
Young's modulus	$E = \sigma/\varepsilon$
Axial deformation	$\delta L = FL/AE$
Shear modulus	$G = \tau/\gamma$
Poisson's ratio	$\nu = -\varepsilon_{\text{lat}}/\varepsilon_{\text{axial}}$
Bulk modulus	$K = \sigma/\varepsilon_v$
Elastic constants	$E = 2G(1+\nu) = 3K(1-2\nu)$
Thermal stress	$\sigma = E\alpha\Delta T$

4.11 Summary

This chapter establishes the language of solid mechanics — stress, strain, and the elastic constants — that underpins all subsequent structural chapters. The key skill is setting up **compatibility** and **equilibrium** equations correctly for statically indeterminate problems such as composite bars and thermally loaded members.

Chapter 5

Torsion

5.1 Assumptions of Simple Torsion Theory

The theory applies to circular cross-sections (solid or hollow) and assumes:

- The material is homogeneous and isotropic
- Plane cross-sections remain plane after twisting
- Shear stress is proportional to shear strain (elastic behaviour)
- The shaft is straight and of uniform cross-section

5.2 The Torsion Formula

The fundamental relationship linking shear stress, torque, geometry, and angle of twist:

$$\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$$

where:

- T = applied torque ($\mathrm{N \cdot m}$)
- J = polar second moment of area (m)
- τ = shear stress at radius r (Pa)
- G = modulus of rigidity (Pa)
- θ = angle of twist (radians)
- L = length of shaft (m)

5.3 Polar Second Moment of Area

For a **solid circular shaft** of diameter d:

$$J = \frac{\pi d^4}{32}$$

For a **hollow circular shaft** with outer diameter d_o and inner diameter d_i:

$$J = \frac{\pi(d_o^4 - d_i^4)}{32}$$

5.4 Shear Stress Distribution

Shear stress varies **linearly** from zero at the centre to a maximum at the outer surface:

$$\tau_{\max} = \frac{T \cdot r}{J} = \frac{16T}{\pi d^3} \quad \text{(solid shaft)}$$

A hollow shaft is more efficient — it carries almost the same torque for less material mass.

5.5 Angle of Twist

The total angle of twist along a shaft of length L:

$$\theta = \frac{TL}{GJ}$$

For a stepped shaft (different diameters along its length), the total twist is the sum of twists in each segment:

$$\theta_{\text{total}} = \sum \frac{T_i L_i}{G J_i}$$

5.6 Power Transmission

The relationship between transmitted power, torque, and rotational speed:

$$P = T\omega = \frac{2\pi N T}{60}$$

where N is rotational speed in rpm and ω is in rad/s. This is used to find the required shaft diameter for a given power and speed.

5.7 Composite (Compound) Shafts

5.7.1 Shafts in Series

The same torque passes through each section; total twist is the sum of individual twists:

$$\theta_{\text{total}} = \frac{TL_1}{GJ_1} + \frac{TL_2}{GJ_2}$$

5.7.2 Shafts in Parallel

Both shafts share the applied torque and twist by the same angle:

$$T = T_1 + T_2 \qquad \theta_1 = \theta_2$$

These two conditions give the equations needed to solve for the torque in each shaft.

5.8 Close-Coiled Helical Springs

A close-coiled helical spring under axial load W is primarily a torsion problem. The wire is subjected to torque $T = WR$, where R is the mean coil radius.

Shear stress in the wire:

$$\tau = \frac{16WR}{\pi d^3}$$

Axial deflection:

$$\delta = \frac{64WR^3n}{Gd^4}$$

Stiffness of the spring:

$$k = \frac{W}{\delta} = \frac{Gd^4}{64R^3n}$$

where d is the wire diameter and n is the number of active coils.

5.9 Summary of Equations

Concept	Equation
Torsion formula	$T/J = \tau/r = G\theta/L$
Polar moment — solid shaft	$J = \pi d^4/32$
Polar moment — hollow shaft	$J = \pi(d_o^4 - d_i^4)/32$
Max shear stress (solid)	$\tau_{\max} = 16T/\pi d^3$
Angle of twist	$\theta = TL/GJ$
Power transmitted	$P = 2\pi NT/60$
Spring deflection	$\delta = 64WR^3n/Gd^4$
Spring stiffness	$k = Gd^4/64R^3n$

5.10 Summary

Torsion analysis always begins with the torsion formula $T/J = \tau/r = G\theta/L$. For composite shaft problems, correctly identifying whether shafts are in **series** (same torque) or **parallel** (same twist) is the critical first step. The close-coiled helical spring is a direct application of torsion theory to a practical engineering component.

Chapter 6

Combined Stress

Chapter 7

Fluid Mechanics

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Attributions

This book incorporates figures, data, and other resources from the following sources:

- Figures in the formulae, the centroids and second moments of area, are adapted from the Wikipedia pages on Centroids and Second moments of area.

Colophon

This book was typeset with Quarto 1.9.36 using XeLaTeX and Latin Modern fonts, with Menlo for code blocks.

Technical Notes

The following tools and technologies were used in the preparation of this book:

- Python: Primary language for computations
- uv: Modern Python package manager and resolver
- Ruff: Fast Python linter and code formatter
- Black: Python code formatter
- Bash: Unix shell for task automation
- macOS: Main development operating system

Appendix A

SI System Common Mistakes

Using the SI system correctly is crucial for clear communication in science and engineering. Below are common mistakes in using the SI system, examples of incorrect usage, and how to correct them.

Table A.1: SI system rules and common mistakes

Concept	**Mistake**	**Correct Usage**	**Notes**
Use of SI Unit Symbols	`m./s`	`m/s`	Use the correct format without additional punctuation.
Spacing Between Value & Unit	`10kg`	`10 kg`	Always leave a space between the number and the unit symbol.
Incorrect Unit Symbols	`sec`, `hrs`, `°K`	`s`, `h`, `K`	Use the proper SI symbols; symbols are case-sensitive.
Abbreviations for Units	`5 kilograms (kgs)`	`5 kilograms (kg)`	Avoid informal abbreviations like "kgs"; adhere to standard symbols.

Concept	Mistake	Correct Usage	Notes
Multiple Units in Expressions	`5 m/s/s`, `5 kg/meter`^{2}	`5 m/s`^{2}, `5 kg/m`^{2}	Use compact, standardized formats for derived units.
Incorrect Use of Prefixes	`0.0001 km`	`100 mm`	Choose prefixes to keep numbers in the range (0.1 x < 1000).
Misplaced Unit Symbols	`5/s`, `kg10`	`5 s`$^{-1}$, `10 kg`	Symbols must follow numerical values, not precede them.
Degrees Celsius vs. Kelvin	`300°K`	`300 K`	Kelvin is written without "degree"
Singular vs. Plural Units	`5 kgs`, `1 meters`	`5 kg`, `1 meter`	Symbols do not pluralize; full unit names follow grammar rules.
Capitalization of Symbols	`Kg`, `S`, `Km`, `MA`	`kg`, `s`, `km`, `mA`	Symbols are case-sensitive; use uppercase only where specified (e.g., `N`, `Pa`).
Capitalization of Unit Names	`Newton`, `Pascal`, `Watt`	`newton`, `pascal`, `watt`	Unit names are lowercase, even if derived from a person's name, unless starting a sentence.
Prefix Capitalization	`MilliMeter`, `MegaWatt`	`millimeter`, `megawatt`	Prefixes are lowercase for (10^{-1}) to (10^{-9}), uppercase for (10^6) and larger (except `k` for kilo).
Formatting in Reports	`5`, `Temperature: 300`	`5 kg`, `Temperature: 300 K`	Always specify units explicitly.

Appendix B

Greek Letters

The following tables present the names of Greek letters and selected symbols commonly used in engineering courses, ensuring precise reference and avoiding reliance on informal descriptors such as "squiggle."

Table B.1: Greek letters.

Lower Case	Upper Case	Name
α	A	alpha
β	B	beta
γ	Γ	gamma
δ	Δ	delta
ϵ	E	epsilon
ζ	Z	zeta
η	E	eta
θ	Θ	theta
ι	I	iota
κ	K	kappa
λ	Λ	lambda
μ	M	mu
ν	N	nu
ξ	Ξ	xi
o	O	omicron
π	Π	pi
ρ	P	rho
σ	Σ	sigma
τ	T	tau
υ	Υ	upsilon
ϕ	Φ	phi
χ	X	chi

Lower Case	Upper Case	Name
ψ	Ψ	psi
ω	Ω	omega

Table B.2: Commonly used symbols in engineering courses.

Symbol	Name	Use	Course
Δ	Delta	Change	Thermodynamics
Δ	Delta	Displacement	Naval Architecture
∇	Nabla	Volume	Naval Architecture
Σ	Sigma	Sum	Thermodynamics, Naval Architecture, Applied Mechanics
σ	Sigma	Stress	Thermodynamics, Applied Mechanics
ϵ	Epsilon	Modulus of elasticity	Thermodynamics, Applied Mechanics
η	Eta	Efficiency	Thermodynamics
μ	Mu	Friction	Thermodynamics, Applied Mechanics
ω	Omega	Angular velocity	Thermodynamics, Applied Mechanics
ρ	Rho	Density	Thermodynamics, Naval Architecture
τ	Tau	Torque	Thermodynamics, Applied Mechanics